National

Operational Excellence Standard (OES) — Bonding Validation Tool (BVT)

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Summary

This standard covers the requirements for the BVT and its operation.

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What's changed in this article

Date	Summary of change
December 3, 2025	Added **Current configuration** section.
September 19, 2025	Added to Celestial.
December 4, 2024	Released document.

At a glance

This article covers the BVT OES.

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BVT requirements and use

This standard will review the requirements for and the use of BVT. BVT is used to assist in validating proper bonding practices at the customer premise as required in OES Premise Bonding requirements.

This document does not substitute for any local, state, or federal code or requirements, but should be used as a supplement to these.

- **Deviations**

    - Basic settings and configurations outlined in this document are the Comcast standard and may not be modified in their implementation. Any deviations from the standards prescribed in this document shall be approved by Comcast Corporate Engineering.
- **Safety**
    - All personnel are required to always comply with Comcast's security and safety regulations.
- **Workmanship**
    - All work performed will be of the highest possible workmanship. All work will adhere to Comcast's Standards, Local, State, and Federal codes / requirements and be fully compliant to any stipulations made by the landlord/agent in control of the property.
- **Work area**
    - The work area must be kept clean and orderly and clear of trash, litter, and debris.

BVT model

The BVT is an application that is available through Tech360 that utilizes Machine Learning (ML) and Artificial Intelligence (AI) to help identify key components required in proper bonding practices.

The BVT is a way to identify proper bonding components and provide feedback to the user in real time. The results will also be stored and can be recalled if needed.

BVT is designed to fit into the technician's current workflow and minimize impacts to time on job. When a technician is validating or completing bonding at a premise, they will utilize their mobile device to take two pictures and submit them for BVT validation.

The tool looks for common bonding hardware, bond tags, and Multimedia over Coax Alliance (MoCA) filter if applicable, needed for proper bonding. If present, then the tool will validate the bond. If not, it will consider the validation a failure and offer an opportunity to resubmit or supply feedback.

BVT is designed for use in residential and commercial environments, excluding any fiber drop type accounts. The tool is not currently designed to identify Tap or ONU bonding.

The tool can be configured to be presented to the technician based on drop type, dwell type, customer type, etc.

There is an option to skip bonding validation on required jobs. Selecting **Skip Bonding Validation** will send an email to the technician's supervisor and manager to make them aware of the action.

When a failure occurs, the result of that failure will be emailed to the technician supervisor and manager.

Reporting is available in OnTRAC under **Bonding Validation Tool Compliance Reports**.

Current configuration

- Drop type — RF
- Dwell type — Residential SFU, Comcast Business, and Comcast Business multi-dwelling unit (MDU)
  - We are currently asking Comcast Business technicians to take images of MDU tap bonding and submit them through the tool to assist in building the MDU tap bonding model.
  - Technicians can change dwell type from Residential SFU to Residential MDU in those cases where they may not be properly marked in the billing system.
- Job types:
  - FDX — Logic will reflect that a MoCA filter should not be present.
  - All IP — Logic will reflect that a MoCA filter should not be present.

BVT use flow

1. The preferred method for utilizing the tool is, after validating, that bonding is correct and all connections/repairs are made. Use the **Contextual Menu** to select **Bonding Validation**.
    - All images should be taken:
        - At 1–2 feet.
        - Components to be detected need to be easily seen and in focus.
        - Facing directly at the components, not at an angle.
        - Well lit, reduce shadows by utilizing the flash.
        - Without obstruction of the items to be validated.
        - Tags should be rotated to be facing forward.
2. Choose either Bond Block, for validating the bond within the house box or Power Bond to validate the connection to the power ground.
3. Follow the on-screen prompts.
4. If a failure occurs, review the failure reasons and either:
    - Correct the failure and retest.
    - Provide feedback and continue with testing the other location.
5. Select the remaining testing location, Bond Block, or Power Bond.
6. Follow on screen prompts.
7. If a failure occurs, review the failure reasons and either:
    - Correct the failure and retest.
    - Provide feedback and continue.
8. If the premise is left in a failed state, the last test run before job closure is a failure, the supervisor will receive an email with the failure reason(s) and a link to the image for verification and feedback to the technician.
9. If testing is not completed by the time the job is closed, the technician will receive a prompt to complete testing before closing the job.

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